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An Information-Theoretic Reformulation of Thermodynamic Gravity (Conjecture)

Paper B (conjecture) — **Draft Status:** 2026-07-29 · Preliminary research · position / research-programme paper **Scope contract:** ../DELIVERY_SCOPE.md §2 (may assert B1–B7; non-claims N1–N9) **Depends on:** Paper A ../01-foundations/PAPER_A_computational_entropy.md · Paper C ../02-computational-models/PAPER_C_decay_algebra.md **Falsifiability verdict:** ../FALSIFIABILITY_L_vs_GR.md **Canonical dynamics:** ../emergent-gravity/master-equation.md · **Newton:** ../emergent-gravity/recoveries/newtonian/README.md

Abstract

Building on the computational-entropy framework of Paper A — where the entropy of a channel’s **output** distribution is a well-defined, Landauer-exact quantity — we ask whether gravity’s known thermodynamic character can be *re-expressed* in this vocabulary. We define a gravitational channel Φ_g and a scalar **computational load** L that reparameterizes proper time, and we present the resulting master equation **honestly as a postulate with a Jacobson-consistent shape**, not as a derivation. Under this postulate the Newtonian weak field is recovered only along an imported route (Clausius → Einstein → Poisson) with one on-shell calibration; we do not derive Newton or Newton’s constant from information alone. We report that our framework and continuum Gravity-from-Entropy reach the same leading Poisson equation (a *weak* agreement forced by both embedding Einstein at low coupling) while diverging at next order. Our central result is therefore **negative-leaning and clarifying**: as constructed the framework is an **information-theoretic reformulation** of thermodynamic gravity, making no sharp prediction distinct from general relativity in any currently accessible regime. We isolate the single term of L — the computational-entropy-production term — where a genuine departure could live, state precisely the missing theorem that would turn it into a prediction, and flag either outcome (reformulation confirmed, or departure found) as acceptable.

1. Position (B1)

This paper is **not** a claim of a new theory of gravity. It is a claim that the *thermodynamic/entropic* content already present in Jacobson’s and Verlinde’s programs can be repackaged in the computational-entropy language of Paper A, and an honest map of exactly how far that repackaging currently reaches. Whether it ever becomes a genuine theory (a falsifiable departure from GR) is **open**; the falsifiability analysis (§6) records the current verdict as *reformulation*.

Every mapping below is tagged **semantic** / **structural** / **constructive**. Type safety is enforced throughout: the load L is a dimensionless scalar; a structure-induced metric G is a tensor; $L \neq G$.

2. The gravitational channel and load (B2)

We model local microstate evolution as a CPTP **gravitational channel** Φ_g ,

$$\rho(\tau + \delta\tau) = \Phi_g[\rho(\tau); g_{\mu\nu}(\rho)], \quad S_c(\Phi_g; \rho) = S(\Phi_g(\rho)),$$

with S_c the von Neumann output entropy (the quantum instance of Paper A’s H_c). The instantaneous processing demand is the **computational load**

$$L(\rho, g) = \beta \frac{E[\rho]}{V\epsilon_0} + \gamma \left| \frac{dS_c}{d\tau} \right|_{\text{reg}} + \delta \frac{S_{\text{boundary}}(\rho)}{S_{\text{BH}}(A)},$$

which clocks proper time via $d\tau = dt/(1 + \alpha L)$, giving the master equation

$$\frac{d\rho}{dt} = \frac{1}{1 + \alpha L(\rho, g)} \mathcal{L}_g[\rho; g_{\mu\nu}(\rho)].$$

Honesty statement (structural). The generator $\mathcal{L}_g$ is *required* to satisfy Clausius $\delta Q = T dS_c$ on local horizons. This is exactly Jacobson’s (1995) condition, from which the Einstein equations follow. The master equation is thus **postulated with the shape that makes it Jacobson-consistent** — it is not independently derived, and on-shell its geometry is GR by construction. What the framework adds beyond GR, if anything, lives entirely in the *clock* L (see §6).

3. Newton via Path J/M only (B3)

We recover the Newtonian weak field along one route, with rigor labels (canonical: [../emerгент-gravity/recoveries/newtonian/README.md](#)):

Step	Content	Rigor
J	Clausius on $\mathcal{L}_g$ Einstein equations	imported theorem (Jacobson) + modeling assumption
—	Einstein weak field $\nabla^2\Phi = 4\pi G\rho_m$	standard GR
M	Load clock $d\tau = dt/(1 + \alpha L)$ matches Newtonian redshift on shell ; $\alpha\beta = 4\pi G/c^4$	calibration

Explicitly not claimed (N5): the withdrawn pointwise identification $\Phi \propto \rho \Rightarrow \nabla^2\Phi \propto \nabla^2\rho$.
Not claimed: a free first-principles value of G , or Newton independent of the Jacobson/Einstein import.

4. Contact with continuum Gravity-from-Entropy (B4)

Comparing our framework to Bianconi’s continuum GfE at low coupling (canonical: [../synthesism6-weak-field-plugtest.md](https://github.com/ianm6/synthesism6-weak-field-plugtest.md)):

Criterion	Outcome
Same leading Poisson $\nabla^2\Phi = 4\pi G\rho_m$	WEAK PASS — both via Einstein/GR
Same derivation mechanism	FAIL
Identifiable $(\alpha_L, \beta_L) = (\alpha_B, \beta_B)$	FAIL / refused
Next-order corrections match	structural FAIL — GfE extras live in the metric EOM $(D_{\mu\nu}, \Lambda_G)$; our γ_L, δ_L live in the load clock unless <i>promoted</i>

The Poisson agreement is **circumstantial co-class with GR**, not evidence that load dynamics equal Bianconi’s relative-entropy equations of motion (non-claim N1).

5. The ACD-EW construction as analogy only (B5)

An earlier warm-up (“Action–Channel Duality, Euclidean”) compares edge-preserving (Perona–Malik) diffusion to isotropic heat on a noisy 1D/2D signal, and finds the edge-preserving reconstructor wins residual on an intermediate time window. We include it **only as an analogy** to Bianconi’s warm-up gradient flow. We note plainly that Perona–Malik edge-preserving diffusion is standard prior art from image processing, that the “field” there is a test signal on a flat lattice (not spacetime), and that the gravity payoff is **analogical, not evidential** (non-claim N7: lattice denoising empirical gravity). This thread is treated as **closed**, not an open crisis.

6. Falsifiability — the central open question (B6)

Since the geometry is GR on-shell, any departure must come from L depending on data not carried by the stress-energy $T_{\mu\nu}$. Term by term (full analysis: [../FALSIFIABILITY_L_vs_GR.md](https://github.com/ianm6/FALSIFIABILITY_L_vs_GR.md)):

1. **Energy term** $\beta E/V\epsilon_0$: a function of $T_{\mu\nu}$, calibrated to GR redshift — **no new prediction**.
2. **Entropy-production term** $\gamma|dS_c/d\tau|$: depends on the *rate of irreversible information processing*, which GR’s proper time does **not** see — **the one candidate departure**.
3. **Boundary term** $\delta S_{\text{boundary}}/S_{\text{BH}}$: only significant near horizons — quantum-gravity regime, not currently testable.

Verdict (recorded): REFORMULATION, with the entropy-production term flagged as the single place a genuine GR departure could live. It is not yet a prediction because (a) degeneracy with an effective stress-energy contribution has not been excluded, and (b) the calibrated magnitude of γ is unknown.

Central conjecture / missing theorem. *Either* exhibit a regime in which $dS_c/d\tau$ is **not** a function of $T_{\mu\nu}$, compute the calibrated γ -residual, and derive a clock-rate prediction distinct from GR ($\rightarrow$ a real experiment: precision timing across strongly irreversible / decoherence-heavy processes at fixed energy density); *or* prove a **degeneracy no-go** showing the three-term load is always reabsorbable into GR + effective source without promotion into the field equations. Either result closes the question and is acceptable.

7. The discrete side is now built — a constructive continuum density (B7)

The reformulation’s weakest point used to be that the discrete export ledger and the continuum load term were joined only semantically, with no constructed continuum object on the discrete side.

That gap is now closed on the discrete side. Paper C ([../02-computational-models/PAPER_C_decay_algebra.md](#)) shows, for the single load slot $L_S = H(X | Y)$, that:

- coupled export is **extensive** (a per-site density exists despite Paper A path-dependence, which saturates to an $O(1)$ boundary);
- a **decay algebra** — a belief transported across the coupling boundary — computes that density, **proved** for 1D ($a = 0.3007568$) and for general window widths (Blackwell representation + reset-regeneration); and
- a **hydrodynamic limit** yields a *constructive continuum entropy-production density field*

$$\sigma(x) = a(\theta(x)) = h_2(\theta(x)) - h_Y(\theta(x)), \quad \text{at rate } O(1/N).$$

So the load’s entropy-production slot now has a **concrete discrete-to-continuum construction** on the computational side. This sharpens the reformulation’s open question from a vague hope to a **well-posed semantic bridge**:

Central open bridge. *Is the classical density $\sigma(x)$ (Paper C) the continuum entropy-production density that appears in the load term $\gamma|dS_c/d\tau|$?*

This is exactly the term (2) candidate of the falsifiability analysis ([../FALSIFIABILITY_L_vs_GR.md](#)). We do **not** assert the identification (non-claim N9, and non-claim 10: σ is a classical density, not the load term). What has changed is that the discrete side of the bridge is no longer aspirational — it is a proved/constructed density field, and the remaining step is precisely the labeled semantic one.

How this could be settled. The concrete experimental path — how to *define* a regime where $dS_c/d\tau$ decouples from $T_{\mu\nu}$, why several regimes are needed (dephasing / erasure / scrambling), and the staged experiments (a computational model witness first, then the calibrated magnitude, then precision-clock tests) — is laid out in [../REGIME_PROGRAM_dSc_decoupling.md](#). Default expectation: a departure in principle (dephasing produces entropy at fixed energy) but a Planck-suppressed magnitude in practice — i.e. reformulation — pending the Stage-1 model witness.

8. Non-claims (frozen)

N1 master eq continuum GfE · N2 $L \not\equiv G$, $S_c \not\equiv \text{Tr } g \ln G^{-1}$ · N5 no pointwise- ρ Newton · N6 $\gamma_L, \delta_L \neq D_{\mu\nu}, \Lambda_G$ · N7 lattice gravity · N8 GfE not on par with GR · N9 no continuum L/G from IDEM · no free G -from-bits. Full list: [../synthesis/CURRENT_CLAIMS.md](#) §3.

Changelog

Date	Entry
2026-07-29	Initial draft: reformulation stance; channel/load postulate; Path J/M; M6 WEAK PASS/FAIL; ACD-EW as analogy; falsifiability verdict + central conjecture; O1 seed.